Solution

FORCE AND PRESSURE -01

Class 08 - Science

1.
(c) Force
Explanation: A push or pull on an object is called force. Force can make stationary body in motion and moving body in rest.

2. **(a)** Friction helps a ship move through water
Explanation: Friction helps a ship move through water

3.
(c) option (i)
Explanation: Pressure is a scalar quantity, not a vector quantity. It has a magnitude but no direction associated with it. Pressure acts in all directions at a point inside a gas. At the surface of a gas, the pressure force acts perpendicular to the surface.

4.
(d) Increased pressure, increases the boiling point
Explanation: The pressure cooker traps that hot air and moisture with the food, which expedites the cooking process. In other words, the moisture surrounding the food itself reaches higher temperatures than it would without the pressure, which speeds up the chemical processes involved in cooking.

5.
(b) parallel
Explanation: parallel

6.
(d) Uniform thrust and different pressure
Explanation: A rectangular block has 6 sides with 4 larger sides and 2 smaller sides as their length is longer than the breadth.

7. **(a)** Use of ball bearings
Explanation: Ball bearings reduces friction of moving parts in a machine. So, sliding friction is replace by rolling friction. Ball bearings are spheres that are held in a track. They are used between a wheel and a fixed axle or between a wheel- axle and a vehicle shaft. They are used to provide better linear motion or rotation around a fixed axis.

8.
(d) Because they are acting on two different bodies
Explanation: Action and reaction forces are equal and opposite to each other but don’t cancel each other because they are acting by two different bodies. Newton's Third Law of motion states that for every action force, there is an equal and opposite reaction force.

9.
(c) 2.5×10^3 N/m^2
Explanation: 2.5×10^3 N/m^2

10.
(c) 10 Pa
Explanation: Here, Force exerted by the box on ground = 20N
Area of the ground the box covers = 2 sq m
Since, Pressure = Force/ Area.
Pressure exerted by the box on the ground = 20N / 2 sq m = 10Pa.

11. **(a)** 20 N
Explanation: Here, Mass of the body = 10kg
Acceleration = 2m/s sq
Force= Mass $\times$ Acceleration =10kg $\times$ 2m/s sq = 20N.

12.
(b) ground
Explanation: According to Newton's third law; every action has an equal and opposite reaction. This is relevant to walking

because when you put your foot on the ground, you are applying a force to it. In doing this, the ground also actually applies an equal force onto your foot, in the opposite direction, pushing you forward.

13.

(d) Height of the column

Explanation: The pressure exerted by water at the bottom of the container depends on the height of the column of water. Pressure of water increases with depth.

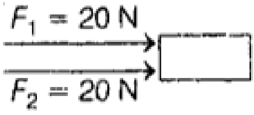
14. **(a)** the effective area of contact between the wheel and axle is reduced

Explanation: the effective area of contact between the wheel and axle is reduced

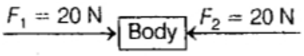
15. **(a)** 40 N and 0

Explanation: When two forces act in the same direction and parallel to each other; then the net force is

F_{net} = 20N + 20 N = 40 N



When two forces act in the opposite direction as shown in the figure.



$$F_{net} = F_1 + F_2$$

Here, F_1 = 20 N

F_2 = - 20N (opposite direction)

= 20 - 20 = 0

16.

(b) Surface irregularities on them

Explanation: Friction between any objects is due to surface irregularities on them. Friction is the resistance to motion of one object moving relative to another. Friction arises due to strong atomic or molecular forces of attraction between the two surfaces at the points of contact. In rough surfaces, the actual area of contact is very small, so the pressure at the points of contact becomes very large. So rough surfaces have more friction than smooth or polished surfaces.

17.

(b) Friction is absent

Explanation: Frictional force always acts in opposite direction of motion to slow down or stopped the body from moving. A body will remain in state of motion on rolling on ground if friction is absent or the body will not move at all as friction is due air also.

18.

(b) lying

Explanation: lying

19.

(b) Unbalanced force

Explanation: Forces that cause a change in the motion of an object are called unbalanced forces. Unbalanced forces are not equal and opposite. Example: In a tug of war the team that pulls harder than the other team wins. This unbalanced force is required to overcome the gravitational force acting on the body.

20.

(c) Gravitational force

Explanation: The force by which earth attracts the other body is called gravitational force. The earth pulls all objects towards itself with a force. Gravitational force depends upon mass of the body and distance from the center of earth. It is a non contact force as the force exerted by the earth can act from a distance.

21.

(c) Option (a)

Explanation: scalar quantity

22.

(c) either in rest or in motion

Explanation: either in rest or in motion

23.

(b) Pull the rope in opposite direction

Explanation: In a game of tug-of war, two teams pull the rope in opposite direction. As the two teams are acting in opposite directions, they will balance each other and there is no net force acting on the rope. As soon as one team applies stronger force and pulls the rope toward itself, the team becomes the winner.

24.

(b) 746 W

Explanation: The electrical equivalent of one horsepower is 746 watts in the International System of Units (SI), and the heat equivalent is 2,545 BTU (British Thermal Units) per hour. Another unit of power is the metric horsepower, which equals 4,500 kilogram-metres per minute (32,549 foot-pounds per minute) or 0.9863 horsepower.

25. **(a)** Better grip of their opponents

Explanation: Kabaddi players rub their hands with soil for a better grip of their opponents. Soil increase the roughness and creates more friction while catching the player firmly such that not to slip.

26. **(a)** 5 N

Explanation: F = ma

$= 0.5 \times 10$

= 5 N

27.

(c) Decrease pressure on road

Explanation: Truck or motor-bus has much wider tyres to decrease the pressure on road because pressure is inversely proportional to area of contact, and this increases the area of contact and reduces pressure.

28.

(b) act on different bodies

Explanation: act on different bodies

29.

(d) 3246.75 Pa

Explanation: Here, Force = 50 N

Radius of circular plate = 7cm

Area of a circle = π r sq= $3.14 \times 7 \times 7$ = 154 cm sq = 154/10000 m sq = 0.0154 m sq.

Pressure = Force/Area = 50N/0.0154 m sq = 3246.75 Pa

30.

(b) 4 kg

Explanation: 4 kg

31. **(a)** Boiling happens at higher temprature

Explanation: Food is cooked more quickly in a pressure cooker because at the higher pressure (1 bar/15 psi), the boiling point of water rises from 100 °C (212 °F) to 121 °C (250 °F). The hotter steam is able to transmit its thermal energy o the food at around 4 times the rate of conventional boiling.

32.

(b) N per sq. m

Explanation: The S.I. unit of pressure is Pascal.One pascal is the pressure that results from the action of 1 N of force acting on a unit area.

33.

(d) A non-contact force is acting on the hall against frictional force.

Explanation: A non-contact force is acting on the ball against frictional force due to which ball is moving on a floor along a straight path with a uniform speed.

34.

(b) Electrostatic force

Explanation: During dry weather, while combing hair, sometimes we experience hair flying apart. The force responsible for this is electrostatic force. The force exerted by a charged body on another charged or uncharged body is known as electrostatic force.

35.

(c) Mass and acceleration

Explanation: Force depends upon mass and acceleration of the body. Actually force acting on a body is directly proportional to product of mass and weight. An object with a larger mass needs a stronger force to be moved along at the same acceleration as an object with a small mass.

36.

(b) 1 kgf = 9.8 N

Explanation: 1 kgf = 9.8 N

37.

(d) Pascal

Explanation: S.I. unit of pressure is Pascal. One Pascal is equal to one Newton per meter square. Larger the force, larger will be the pressure exerted.

38.

(d) increase friction force acts on it

Explanation: increase friction force acts on it

39.

(c) the table exerts a frictional force on it

Explanation: the table exerts a frictional force on it

40.

(d) Decreases

Explanation: At high altitudes the atmospheric pressure decreases, as density of air decrease as we move upward. Due to the difference in the atmospheric pressure and the internal pressure developed by the fluids inside the body cells sometimes popping of ears occurs at high altitudes.

41.

(c) Acceleration

Explanation: The rate of change in velocity is called acceleration. Acceleration is a vector quantity and has both magnitude and direction. An object's acceleration is the net result of all forces acting on the object.

42.

(b) 1000 kg

Explanation: The weight of air in a column of the height of the atmosphere and area 10cm $\times$ 10cm is equal to 1000 kg. Atmospheric pressure is the weight exerted by the overhead atmosphere on a unit area of surface.

43.

(b) Equal in all direction

Explanation: The lateral pressure exerted by liquid at same height is equal in all direction and pressure increases with depth.

44.

(c) Increase in depth

Explanation: Pressure exerted by liquid increases with an increase in depth. As the depth of liquid increases, the weight of liquid column pushing down from above increases, and hence the pressure also increases. To protect this pressure, sea divers uses a special suit.

45.

(c) mass

Explanation: mass

46. **(a)** When lift falls freely downward

Explanation: A man is standing on a weighing machine placed in a lift. The reading of machine will be minimum when machine will move when lift falls freely downward.

47.

(c) 1.01×10^5 N/m^2

Explanation: 1.01×10^5 N/m^2

48.

(d) Force of friction

Explanation: When the applied force is in contact with an object, then this is called contact force. Among the given options, the force of friction is a contact force because the force of friction arises due to contact between surfaces.
Magnetic force, the force of gravity, and electrostatic force are non-contact forces.

49. **(a)** water reduces the friction between the road and the tyres

Explanation: water reduces the friction between the road and the tyres

50.

(c) Equal to the atmospheric pressure

Explanation: The pressure inside our body is equal to atmospheric pressure due to which we do not get crushed under the pressure of the atmosphere. The internal pressure developed by the fluids inside the body cells and the atmospheric pressure are equal. At high altitudes where the atmospheric pressure is less then the pressure inside of our body, we feel popping in the air.

51.

(b) Non-contact force

Explanation: Muscular force is example of contact force because the interacting bodies are in contact to each other. Magnetic force is an example of non-contact force as the interacting bodies need not be in contact with each other.

52.

(b) Provide better grips with ground

Explanation: The threaded tyres of truck, cars and buses provide better grips with ground because threads in tyres creates more irregularity to the surface of the road and causes greater frictional force to act and form better grips.

53. **(a)** 1 Pa

Explanation: The S.I. unit of pressure is Pascal. One Pascal is equal to the pressure that results from the action of one Newton of force on one meter square area.
So,1Pascal = 1 N / 1 sq m.

54.

(d) N-m^{-1}

Explanation: The SI unit of pressure is pascal. The pressure is given by
$p = \frac{\text{Force}}{\text{Area}} = \frac{\text{N}}{\text{m}^2}$

In C.G.S system the unit of pressure is dyne/cm^2.

Thus, N-m^{-1} is not a unit of pressure.

55.

(d) On momentum conservation

Explanation: The rocket works because of the law of conservation of linear momentum. The law of conservation of linear momentum is very important in physics. Momentum is defined as the mass of an object times its velocity.

56.

(c) either a magnetic or an electrostatic force

Explanation: When two objects repel each other, this repulsion could be due to either magnetic or electrostatic force. In electrostatic force, similar charges may repel each other. In a magnetic force, similar poles may repel each other.

57.

(b) decreased

Explanation: decreased

58.

(c) Muscular

Explanation: Muscular force is applied to change the shape of a ball of dough to a chapatti. This force leads to the change in shape and size of chapatti.

59. **(a)** the earth attracts it

Explanation: the earth attracts it

60.

(c) contact force

Explanation: contact force

61.

(c) Inversely proportion to area of contact

Explanation: Pressure is inversely proportional to area of contact and directly proportional to force acting on per unit area. Increased area of contact results in applying bigger force which results in lower pressure and less area of contact results in less force which in turn results in high pressure.

62.

(b) muscular

Explanation: muscular

63.

(b) Dyne

Explanation: Dyne

64.

(c) Decrease pressure

Explanation: School bags have broader strips to decrease pressure that is applied on the shoulder. The load remains the same but the area increases. Since pressure is inversely proportional to area, the greater the area, the lesser the pressure, and the pressure on the shoulder reduces in broad straps.

65.

(d) Action of force

Explanation: The motion imparted to object is due to the action of force. If force acts on the object along the direction of motion, its speed increases. If force is applied on a direction opposite to the direction of motion, its speed decreases.

66. **(a)** 20 N

Explanation: Since the two forces are acting in opposite directions, the net force is the difference between the two forces. The net force will act in the direction of the larger force.

Force exerted by the first boy = 40N

Force exerted by the second boy = 60N

The resultant force exerted by two boys on same object will be : Force exerted by the second boy (60N) - Force exerted by the first boy(40N) = 20N

67.

(b) While standing on one leg

Explanation: Any person standing on one leg creates maximum pressure on earth.

68.

(d) Less friction

Explanation: A motorcycle with worn out threads on tyres is more likely to skid due to less friction on slippery road. Tyres have grooves to increase the friction, if these grooves are worn out the grip loses and the friction between the tyre and the surface of the road decreases. There will be less friction between the motorcycle and the road which is most likely to skid.

69.

(c) Overcome frictional force

Explanation: Frictional force always works in opposite direction of motion. A body moving on leveled grounds need to apply constant force to overcome frictional force provided by ground.

70.

(b) Speed of the body increases

Explanation: If the force applied on a moving body is in the same direction of motion than the speed of the body increases. As both the forces are acting in the same direction, they add up to make a bigger force. This large force makes the speed increases. And if force acts in the opposite direction, speed decreases.

71.

(c) in the bottom of the bottle

Explanation: in the bottom of the bottle

72.

(d) Flat shoes have larger surface area

Explanation: It is easy to walk on the sand with flat shoes because flat shoes have a larger surface areas. As pressure is inversely proportional to area. So, a larger surface area will produce less pressure on sand and it will become easy to walk.

73.

(c) 2.5kg

Explanation: Here, force = 5 N and acceleration =2m/s . Force = Mass $\times$ Acceleration, Mass = Force/Acceleration. So, Mass = 5 N/2m/s = 2.5 Kg

74.

(d) Drag

Explanation: The frictional force exerted by fluids is also called drag. Water and other liquids exert force of friction when objects move through them. The friction force exerted by fluids depends on its speed with respect to fluid and also on the shape of the object and the nature of the fluid.

75.

(d) Change in mass

Explanation: Force can change the state of motion, the shape, and size of the object, and change the state of motion but can never change the mass of the body. Force is the product of mass and acceleration.